

An algebraic derivation of the Mundell-Fleming Model

Alejandro Melo Ponce

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1 Key assumptions

- Domestic real interest rate is equal to the world real interest rate: $r = r^*$.
- Perfect Capital Mobility
- In the short-run the price level P is fixed.

2 The Goods Market and the IS* curve

Planned expenditures can be described by the following equation:

$$Y^d = C(Y - T) + I(r) + G + NX(e), \quad (1)$$

where Y^d is planned expenditures; Y is actual expenditures; C is consumption which is a function of disposable income $Y - T$, where T represents taxes levied by the government; I is investment in capital goods; G is government expenditures; NX are net exports where e is the nominal exchange rate. Recall that the nominal exchange rate e is related to the real exchange ϵ via the equation $\epsilon = \frac{eP}{P^*}$, where P is the domestic price level and P^* is the world price level.

For the aggregate level of consumption, we will assume that it takes the following form:

$$C(Y - T) = C_0 + \text{mpc}(Y - T), \quad (2)$$

where $C_0 \geq 0$ is an autonomous level of consumption and $0 < \text{mpc} < 1$ is the *marginal propensity to consume*.

For aggregate capital investment demand, we will assume that it takes the following form:

$$I = I_0 - \delta r, \quad (3)$$

where $I_0 \geq 0$ is an autonomous level of consumption and $\delta > 0$ is the sensitivity of investment to the real interest rate.

Finally, assume that that net exports take the following form:

$$NX(e) = NX_0 - \gamma e - \text{mpi}Y, \quad (4)$$

where NX_0 is an autonomous level of net exports,¹ $\gamma > 0$ is the sensitivity of net exports to the interest rate and mpi is the marginal propensity to import.

¹ $NX_0 = X_0 - I\bar{M}$, where X_0 and $I\bar{M}$ are autonomous levels of exports and imports respectively. Thus, NX_0 can be positive, negative or zero.

Thus, combining equations (1), (2), (3) and (4) we have that planned expenditures can be written as

$$Y^d = C_0 + \text{mpc}(Y - T) + I_0 - \delta r + G + NX_0 - \gamma e - \text{mpi}Y,$$

and collecting terms we get

$$Y^d = \underbrace{[C_0 - \text{mpc}T + I_0 - \delta r + G + NX_0]}_{A(C_0, T, I_0, r, G, NX_0)} - \gamma e + (\text{mpc} - \text{mpi})Y, \quad (5)$$

where the $A(C_0, T, I_0, r, G, NX_0)$ term collects the autonomous components.

The equilibrium conditions for the goods market, together with the condition $r = r^*$ yields:

$$Y = Y^d, \quad (6)$$

$$Y = A(C_0, T, I_0, r^*, G, NX_0) - \gamma e + (\text{mpc} - \text{mpi})Y, \quad (7)$$

solving for Y yields the IS* curve:

$$Y = \frac{A(C_0, T, I_0, r^*, G, NX_0)}{1 - \text{mpc} + \text{mpi}} - \frac{\gamma}{1 - \text{mpc} + \text{mpi}}e \quad (\text{IS}^*)$$

Notice that the term $A(C_0, T, I_0, r^*, G, NX_0)$ acts as a shifter of the IS* curve.

3 The Liquidity Market and the LM* curve

Consider the following functional form for the demand for real money balances $L^d(r, Y)$:

$$L^d(r, Y) = \alpha Y - \beta(r^* + \pi^e), \quad (8)$$

where $\alpha > 0$ is the sensitivity of the demand for real money balances to aggregate income and $\beta > 0$ is the sensitivity of the demand for real money balances to the nominal interest rate²

Thus, the equilibrium condition for the money market, together with the condition $r = r^*$ yields:

$$\frac{M^s}{P} = L^d(r^*, Y) \quad (9)$$

$$\frac{M^s}{P} = \alpha Y - \beta(r^* + \pi^e) \quad (10)$$

$$Y = \frac{M^s}{\alpha P} + \frac{\beta}{\alpha}(r^* + \pi^e).. \quad (\text{LM}^*)$$

Notice how monetary policy M^s and inflation expectations π^e shift the LM* curve. Most importantly, notice that the LM* curve does not depend on the exchange rate.

4 Putting all the pieces together

The equations

$$Y = C(Y - T) + I(r^*) + G + NX(e) \quad (\text{IS}^*)$$

$$\frac{M^s}{P} = L(r^*, Y), \quad (\text{LM}^*)$$

²Recall that the Fisher equation tells us that $i = r^* + \pi^e$.

characterize the equilibrium aggregate income Y^* and nominal exchange rate e^* that get the money market and goods market in equilibrium at the same time given the world interest rate r^* . The endogenous variables in the model are Y and e and the exogenous variables are T, G, r^* and all the autonomous components of the individual aggregate functions.

5 Example

Consider the following information:

$$\begin{array}{ll} C = 0.5(Y - T) & \frac{M^s}{P} = 0.5Y - 500r \\ T = 1000 & M^s = 1000 \\ I = 1500 - 250r & G = 1500 \\ NX = 1000 - 250e & r^* = 2, P = 1. \end{array}$$

We will calculate the short-run equilibrium.

IS*

$$\begin{aligned} Y &= 0.5(Y - T) + 1500 - 250r^* + 1500 + 1000 - 250e \\ Y &= 0.5(Y - 1000) + 1500 - 250(2) + 1500 + 1000 - 250e \\ Y &= 0.5Y - 500 + 1500 - 500 + 1500 + 1000 - 250e \\ Y(1 - .5) &= 3000 - 250e \\ Y &= 6000 - 500e \rightarrow \text{IS}^*. \end{aligned}$$

LM*

$$\begin{aligned} \frac{M^s}{P} &= 0.5Y - 500r^* \\ 1000 &= 0.5Y - 500(r^*) \\ 1000 &= 0.5Y - 1000 \\ 2000 &= 0.5Y \\ Y &= 4000 \rightarrow \text{LM}^*. \end{aligned}$$

Thus, we have that:

$$\begin{aligned} LM^* &= IS^* \\ 4000 &= 6000 - 500e \\ 500e &= 2000 \\ e^* &= 4, Y^* = 4000. \end{aligned}$$

Thus, $C = 0.5(4000 - 1000) = 1500$, $I = 1500 - 250(2) = 1000$, $NX = 1000 - 250(4) = 0$.

Recall that we have three types of savings that make up national savings, foreign, private and public:

$$\begin{aligned} S_{foreign} &= -NX, \\ S_{private} &= (Y - T) - C, \\ S_{public} &= T - G. \end{aligned}$$

Thus $S_{foreign} = 0$, $S_{private} = 1500$, and $S_{public} = -500$. Thus $S_{national} = S_{foreign} + S_{private} + S_{public} = 1000$.